

Quantitative relative balance at the endpoint

Asymptotic research note; independently reviewed

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Assume $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ are nondecreasing, the positive row weights w sum to one, and

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

Write $u_i = 1 - p_i$ and $s = \sum_i u_i$. Constants below are absolute, independent of the number of rows, weights, and chosen column.

Theorem 1. *For all $m \geq 2$ and every row,*

$$\max_i (1 - p_i) \leq \frac{C(s + m^{-1/3})}{m}. \quad (1)$$

This strengthens the previously proved bound $C(s + 1)/m$. It is a structural tensor assertion; no new per-die exponent is claimed in this note.

Fix a column i , put $M = m - 1$, and write $s_- = \sum_{j \neq i} u_j$. Set

$$\ell = \lfloor M/4 \rfloor, \quad N = M - \ell, \quad \alpha = \ell/M, \quad x = \alpha s_-, \quad \theta = 1/16.$$

We work with sufficiently large M ; smaller values are covered by increasing C , since $u_i \leq s$. For a uniformly selected ℓ -subset B of the columns other than i , put $A_B = \prod_{j \in B} p_j$ and $\bar{A} = \mathbb{E}_B A_B$. There are two probability measures on the original rows:

$$\mu_0 = (\ell + 1)w\bar{A}, \quad \mu_1 = (\ell + 1)(\ell + 2)wu_i\bar{A}. \quad (2)$$

Their exact density relation, including zero entries, is

$$d\mu_1 = (\ell + 2)u_i d\mu_0. \quad (3)$$

The corresponding scalar probability densities are

$$\begin{aligned} b_0(z) &= \frac{\ell+1}{\ell} (1-z/\ell)^\ell \mathbf{1}_{[0,\ell]}(z), \\ b_1(z) &= \frac{(\ell+1)(\ell+2)}{\ell^2} z(1-z/\ell)^\ell \mathbf{1}_{[0,\ell]}(z). \end{aligned}$$

Let β_a denote integration against b_a .

Lemma 1 (The marked comparison). *For $a = 0, 1$, $1 \leq D \leq \sqrt{M}$, and $Q_b(z) = \sum_{j=0}^D c_{bj} L_j(\theta z)$ with $B_b = \sum_j |c_{bj}|$, one has*

$$|\mathbb{E}_{\mu_a} Q_1(x) Q_2(x) - \beta_a(Q_1 Q_2)| \leq C B_1 B_2 \left(\frac{D}{M} + e^{-cM} \right). \quad (4)$$

The constants are uniform in i .

Proof. For $a = 0$, this is the scalar endpoint comparison in `../power_per_die_lower_bound.tex`, applied to the M columns other than i . They still satisfy exactly the same uniform mixed-moment identities with the original row weights. Thus its application is not circular.

We detail the changes needed for $a = 1$. The finite global balance bound for all m columns gives

$$u_i \leq \frac{256(s_- + u_i + 1)}{M+1} \leq \frac{C(s_- + 1)}{M} \quad (5)$$

for large M , after moving the u_i term to the left. The finite balance and row-mass bounds applied to the remaining M columns give

$$\max_{j \neq i} u_j \leq C(s_- + 1)/M, \quad \bar{A} \leq e^{-\alpha s_-}, \quad \mathbb{E}_w(s_- + 1)^k e^{-cs_-} \leq C_c^{k+1} k! / M. \quad (6)$$

In particular the marked normalization obeys, for integers $k \geq 0$,

$$(\ell+1)(\ell+2) \mathbb{E}_w[u_i(s_- + 1)^k e^{-cs_-}] \leq C_c^{k+2} (k+1)!. \quad (7)$$

For $P = Q_1 Q_2$, multilinearity and the mixed moments give the exact identity

$$(\ell+1)(\ell+2) \mathbb{E}_w \mathbb{E}_B[u_i A_B \text{Pol}_N(P)((\ell u_j)_{j \notin B \cup \{i\}})] = \beta_1(P). \quad (8)$$

Indeed its scalar integral is $(\ell+1)(\ell+2) \int_0^1 (1-t) t^\ell P(\ell(1-t)) dt$. The supports of the mark, forced successes, and polarized factors are disjoint.

The pointwise polarization argument from the cited comparison proof is unchanged. If v is the mean of the remaining ℓu_j , its centered variance is at most $CN(s_- + 1)v$. The off-diagonal Laguerre derivative estimate cancels the factor v in the centered coefficient estimate, giving pointwise error at most

$$B_1 B_2 e^{\theta v} \sum_{k=2}^{2D} \frac{(C\sqrt{Dk}(s_- + 1)/M)^k}{k!}.$$

Average with $u_i A_B$, apply (7), and include its normalization. The resulting sum is at most

$$CB_1 B_2 \sum_{k=2}^{2D} (C\sqrt{D/M})^k \frac{k^{k/2}(\lceil k/2 \rceil + 1)!}{k!} \leq CB_1 B_2 D/M. \quad (9)$$

The extra factorial factor compared with the unmarked argument is only polynomial in k ; the geometric ratio tends to zero uniformly for $D \leq \sqrt{M}$.

To replace v by x , use the same success-weighted subset law. On $s_- \leq M/256$ its bias and second moment satisfy

$$0 \leq \mathbb{E}_B(v - x) \leq Cs_-(s_- + 1)/M, \\ \mathbb{E}_B(v - x)^2 \leq C \left(\frac{s_-(s_- + 1)}{M} + \frac{s_-^2(s_- + 1)^2}{M^2} \right).$$

These depend only on the remaining columns. The singular Taylor remainder established in `weighted_success_subset_moments.tex` bounds the pointwise error by

$$CB_1 B_2 e^{\theta \alpha s_-/(1-\alpha)} \left[\frac{\sqrt{D}\sqrt{s_-}(s_- + 1)}{M} + \frac{D(s_- + 1)}{M} + \frac{Ds_-(s_- + 1)^2}{M^2} \right].$$

Multiplication by $u_i \bar{A}$ and (7) bound its normalized integral by $CB_1 B_2 D/M$. On $s_- > M/256$, the original uniform-subset Cauchy–Schwarz bound leaves an exponentially decreasing factor; the additional normalization $O(M^2)$ and $u_i \leq 1$ are absorbed in e^{-cM} . This proves (4). $\square$

Lemma 2 (Laplace tests without a logarithmic loss). *Uniformly for $1/2 \leq t \leq M^{1/3}$ and $a = 0, 1$,*

$$|\mathbb{E}_{\mu_a} e^{-tx} - \beta_a(e^{-tz})| \leq Ct/M + C \exp(-c\sqrt{M}/t). \quad (10)$$

Consequently, for all sufficiently large M and $1 \leq t \leq M^{1/3}$,

$$c/t \leq \mathbb{E}_{\mu_0} e^{-tx} \leq C/t, \quad \mathbb{E}_{\mu_1} e^{-tx} \leq C/t^2. \quad (11)$$

The upper bound for μ_0 also holds when $t \geq 1/2$.

Proof. The Laguerre generating function gives

$$e^{-tz/2} = (1-r) \sum_{j=0}^{\infty} r^j L_j(\theta z), \quad r = \frac{t}{t+2\theta}.$$

Truncate at $D = \lfloor \sqrt{M} \rfloor$, calling the resulting polynomial Q_D . Since $|L_j(\theta z)| \leq e^{\theta z/2}$,

$$|Q_D(z) - e^{-tz/2}| \leq r^{D+1} e^{\theta z/2}, \quad |Q_D(z)^2 - e^{-tz}| \leq 2r^{D+1} e^{\theta z}. \quad (12)$$

Both μ_a and β_a integrate $e^{\theta z}$ to at most an absolute constant: this follows from (6)–(7) and $\bar{A} \leq e^{-\alpha s_-}$, or directly from their scalar densities.

Apply (4) separately to $L_j L_k$, using degree $\max(j, k)$; the constant product has zero error since both measures have mass one. For $c_j = (1-r)r^j$, this yields

$$\begin{aligned} |\mathbb{E}_{\mu_a} Q_D(x)^2 - \beta_a(Q_D^2)| &\leq \frac{C}{M} \sum_{j,k \leq D} c_j c_k \max(j, k) + C e^{-cM} \\ &\leq \frac{2C}{M} \sum_{j \geq 0} j c_j + C e^{-cM} \\ &\leq Ct/M + C e^{-cM}. \end{aligned}$$

Furthermore $r^{D+1} \leq e^{-c\sqrt{M}/t}$ for $t \geq 1/2$. Together with (12) this proves (10). The scalar bounds are immediate from $b_0(z) \leq 2e^{-z}$, $b_1(z) \leq Cze^{-z}$, and $b_0(z) \geq 1/2$ on $0 \leq z \leq 1/2$. For $t \leq M^{1/3}$, the error in (10) is negligible compared with $1/t$, and at most C/t^2 . This proves (11). $\square$

Lemma 3 (An interval with controlled density ratio). *There are absolute constants $0 < a < b$ and $c, C > 0$ such that, for all large M and $1 \leq t \leq M^{1/3}$,*

$$\mu_0(a/t < x < b/t) \geq c/t, \quad \mu_1(x \leq b/t) \leq C/t^2. \quad (13)$$

Proof. First the unmarked comparison (4) implies

$$\mu_0(x \leq y) \leq C(y + M^{-1/2}) \quad (y \geq 0). \quad (14)$$

To see this directly, normalize $T_D(\theta z) = \sum_{j < D} (1 - j/D) L_j(\theta z)$ by its coefficient sum $(D+1)/2$. Its square is bounded below by an absolute positive constant on $z \leq c_0/D$, has scalar expectation $O(1/D)$, and has comparison error $O(D/M)$. Choose D comparable to $\min(\sqrt{M}, c_0/y)$; larger y are covered by probability at most one. This proves (14) using only the M -column unmarked tensor.

By (11), $\mathbb{E}_{\mu_0} e^{-tx} \geq c_1/t$. Choose $a > 0$ sufficiently small that the bound (14) for $y = a/t$ is at most $c_1/(4t)$ for large M . This is uniform because $tM^{-1/2} \leq M^{-1/6}$. For the upper tail,

$$\mathbb{E}_{\mu_0}[e^{-tx} \mathbf{1}_{x \geq b/t}] \leq e^{-b/2} \mathbb{E}_{\mu_0} e^{-tx/2} \leq C e^{-b/2}/t.$$

Choose b large enough to make this at most $c_1/(4t)$. The remaining interval therefore carries at least $c_1/(2t)$ weighted mass, hence at least that much probability. The marked bound follows from (11) and $\mathbf{1}_{x \leq b/t} \leq e^b e^{-tx}$. $\square$

Proof of Theorem 1. Fix any row q . If $x(q) \leq a/(2t)$, every row in the interval of (13) has strictly larger x , and hence lies strictly before q . Monotonicity gives $u_i \geq u_i(q)$ throughout that interval. The density relation (3) and (13) give

$$(\ell + 2)u_i(q) \frac{c}{t} \leq \mu_1(a/t < x < b/t) \leq C/t^2, \quad u_i(q) \leq C/(Mt). \quad (15)$$

If $x(q) \leq a/2$, take $t = \min(M^{1/3}, a/(2x(q)))$, with the second entry interpreted as $+\infty$ at $x(q) = 0$. This belongs to $[1, M^{1/3}]$, and (15) gives

$$Mu_i(q) \leq C(x(q) + M^{-1/3}) \leq C(s(q) + M^{-1/3}).$$

If $x(q) > a/2$, then $s(q) \geq s_-(q) \geq a/(2\alpha)$ is bounded below by a positive constant. The old global bound $u_i \leq 256(s+1)/m$ is therefore already $u_i \leq Cs/m$. The column i was arbitrary, and $M = m - 1$ is comparable to m . This proves (1), including rows containing zero entries. $\square$